Exploring Limits

Pre-Lab

We will soon begin to explore the notion of a *derivative* of a function. To do so, we will first need to understand the concept of a *limit*. In this lab we will get going on that.

Rather than attempting to present a technical definition of a limit, we will plunge straight in and play with a few of them. The notion to keep in mind is:

Definition (vague): The *limit* of a function $f(x)$ as x approaches some value a on the x-axis is the y-value we *expect* the function to take at that value of x, even if the actual value is different.

Falling into a River Watch the video linked below, making sure you understand the ideas presented in it. There will be questions on the content in the Gradescope assignment for this pre-lab.

Video: Limits Lab

Some Spreadsheet Work Open a blank Google Sheets spreadsheet, and create three tabs in it. Use each of your tabs to complete the following tables, computing the values of each function at the x values given:

1. $f(x) = \dfrac{(x-7)(x-2)}{x^2+1}$

x	$f(x)$	x	$f(x)$
0.1		-0.1	
0.01		-0.01	
0.001		-0.001	
0.0001		-0.0001	

(Hint: if you do this right, the last entry in the second column should be 13.9991 and the last entry in the fourth column should be 14.0009. If you didn't get that, check your formulas and make sure you are using parentheses correctly.)

2. $f(x) = \dfrac{2x^2 - x - 1}{x - 1}$

x	$f(x)$	x	$f(x)$
1.1		0.9	
1.01		0.99	
1.001		0.999	
1.0001		0.9999	

(Hint: if you do this right, the last entry in the second column should be 3.0002 and the last entry in the fourth column should be 2.9998. If you didn't get that, check your formulas and make sure you are using parentheses correctly.)

3. $h(x) = \dfrac{x + 10000}{9999.9999}$

x	$h(x)$	x	$h(x)$
0.1		-0.1	
0.01		-0.01	
0.001		-0.001	
0.0001		-0.0001	

(Hint: if you do this right, the last entry in the second column should be 1.00000002 and the last entry in the fourth column should be 1. If you didn't get that, check your formulas and make sure you are using parentheses correctly.)

Go do the Gradescope assignment for this pre-lab!

Part I: Introduction to Limits

1. Recall that $\lim_{x\to 0} \dfrac{(x-7)(x-2)}{x^2+1}$ is the value *we expect* this function to take at $x = 0$, regardless of the actual value of the function there. Looking at the first table in your spreadsheet, what value do you expect the limit to be?

2. Now calculate $f(0)$. How is this related to your answer to question 1?

In this case, what we expected to get (the limit) was the same as the actual value of the function at that point. In the video in the pre-lab, this was not the case. There, we expected to be at the top of the bridge, but instead we fell through to the river.

3. Fill in the blank: if we get what we expect, then at that point, the function is _____. (Hint: starts with a 'c'.)

Part II: Indeterminant Limits

With the function in Part 1, there was no difference between computing the limit as $x \to 0$ and evaluating the function at $x = 0$. That is, we expected to get a certain value as we approached $x = 0$ from either side (using the table), and that value matched the actual value of the function at $x = 0$.

However, this is not always the case. If it was, the concept of a limit would not be a very interesting one!

Suppose that $f(x) = \dfrac{2x^2 - x - 1}{x - 1}$ (this is the table in the second tab of your spreadsheet.)

4. What value do you get when you try to calculate $f(1)$?

Indeterminate Values

Expressions that yield $\frac{0}{0}$ are referred to as *indeterminate values* and require more detailed analysis to determine their value (if their value even exists). They will come up often in Math 105L and 106L.

5. Using the table in the second tab of your spreadsheet, what does $\lim_{x\to 1} f(x)$ appear to be?

6. Note that unlike the function in Question 1, this function is not continuous. To see this, click on this link to Wolfram Alpha, and type in the following (or just copy and paste it):

   ```
   discontinuities of (2x^2-x-1)/(x-1)
   ```

   Take a screenshot of the top graph that appears in Wolfram Alpha, and paste it into your Google doc.

   Looking at your graph, is $f(1)$ defined? What is the domain of $f(x)$?

The point $(1, 3)$ is called a "hole" in the graph of $f(x)$ because $f(x)$ seems to run through the point $(1, 3)$, yet it is not defined there. The function is perfectly linear for all other values of x, but seems to just skip over the point $(1, 3)$.

7. What linear function does $f(x)$ resemble for $x \neq 1$? To find out, factor the numerator of $f(x)$ and then simplify. Call this simplified expression $g(x)$. You should find that has a simple linear form.

8. Use Wolfram Alpha to draw the graph of $g(x) = 2x + 1$:

   ```
   2x+1 from x =-0.5 to x=1.5
   ```

   Paste a screenshot next to your graph of $f(x)$ in your Google doc.

9. What is $g(1)$? What is the relationship between $\lim_{x \to 1} f(x)$ and $g(1)$?

   Are $f(x)$ and $g(x)$ the same function?

10. From your observations above, describe how to locate a 'hole' in the graph of a function using the formula for the function.

Estimating limits vs Computing Limits

The table method is a good way to *estimate* limits. The table method is not always reliable, as we will see in the next part of this lab.

What you did in the last few questions is a way to *compute* limits. Computation will give exact answers, while estimation may only give approximate answers.

Part III: Estimating Doesn't Always Work!

We will now consider the function $h(x) = \dfrac{x + 10000}{9999.9999}$ and examine the limit $\lim_{x \to 0} h(x)$. This is the third tab of your spreadsheet.

11. Based on your table, what does $\lim_{x \to 0} h(x)$ appear to be?

12. Calculate $h(0)$.

13. What can you say about the weaknesses of the table method we've been using?

Part IV: Limits at Infinity

14. We can also find limits "at infinity." To see what this means, graph the function

$$f(x) = \frac{2x^2 + 7}{x^2 - x + 1}$$

using Geogebra (or Desmos) and set the x-axis to -100 to 100 and the $y - axis$ from -1 to 15[1]. Take a screenshot and paste it in your Google doc.

15. What does the graph do towards the edges of the screen? That is, what number does $f(x)$ seem to approach as x gets very large in the positive or negative direction?

These numbers are the limits "at infinity" and "at negative infinity" and is denoted by

$$\lim_{x \to +\infty} f(x) \text{ and } \lim_{x \to -\infty} f(x).$$

16. What does the value of this limit seem to be?

17. Describe the relationship between limits at infinity and horizontal asymptotes.

Part IV: Another Indeterminate Limit

Consider the function $f(x) = x^{\frac{1}{x-1}}$.

18. At what value of x is this function undefined?

19. Previously, we saw that the form $\frac{0}{0}$ is indeterminate. what is the form of this function at the discontinuity? There are a number of indeterminate forms, two of which $\frac{0}{0}$ and this one.

20. Use Wolfram Alpha to look at the discontinuity in this in the graph of this function like you did in Question 6. Screenshot the graph that appears and paste it in your Google doc.

21. If you type

```
limit of x^(1/(x-1)) as x approaches 1
```

into Wolfram Alpha, it will tell you what the limit is. What is the limit? Have you encountered this number before? It will come up a lot a bit later in the class!

[1] Geogebra: Gear icon in the top right → Graphics → Dimensions. Desmos: Wrench icon in the top right.